

Que:1 [10 marks] Define ANY TEN from the following terms with clear mathematical equation:

- (a) axiomatic probability, (b) mutually exclusive events, (c) conditional probability, (d) statistically independent events, (e) a posteriori probability, (f) random variable, (g) probability density function, (h) cumulative distribution function, (i) expectation, (j) variance, (k) standard deviation, (l) covariance, (m) correlation, (n) n -variate Gaussian probability density function, (o) Cauchy-Schwarz inequality, (p) Chebyshev inequality, (q) Markov inequality, (r) random process, (s) autocorrelation function, (t) strict sense stationary random process, (u) wide sense stationary random process, (v) white noise process, (w) Gaussian process, (x) ergodic random process, (y) AWGN, and (z) IID.

Que:2 [5 marks] We wish to perform polynomial regression to find a parabola $y = ax^2 + bx + c$ going through following pairs of (x, y) : $(1, -1)$, $(0, 1)$, and $(2, 7)$. Now, (a) write the Vandermonde system to be solved to estimate the unknowns, (b) solve the system for the unknowns (a, b, c) , and (c) write the equation of polynomial y in terms of the estimated values of (a, b, c) .

Que:3 [5 marks] Show the equivalence among the solutions for the minimum mean-squared error estimation (MMSE), the least-squares estimation (LSE), and maximum likelihood estimation (MLE). Write down the problem set-up, mention assumptions, derive necessary formulae, and show geometry of the problem and solution.

-OR-

Que:3 (OR) [5 marks] Let $X = (X_1, X_2)$ consist of two zero-mean, unit variance, uncorrelated random variables. Find the matrix A such that $Y = AX$ has covariance matrix $C_Y = \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix}$. See that it is generating *correlated* random variables *unlike* principal component analysis (PCA). It is an example of data generating processes. **Hint:** Desired matrix $A = P\sqrt{\Lambda}$, where columns of P matrix consist of eigenvectors of C_Y , and Λ is a diagonal matrix of eigenvalues of C_Y .

Que:4 [5 marks] If $Z = X + Y$, where X and Y are independent random variables with $f_X(x) \sim U(-10, 10)$ and $f_Y(y) \sim U(0, 1)$, where $U(\cdot)$ is uniform distribution. Determine minimum mean-squared error (MMSE) estimate of Z in terms of X .

Que:5 On a sample of measurements for the population mean, the readings are as follows:

$[2.5, 7.4, 8.0, 4.5, 7.4, 9.2]$

- Compute the five-point summaries and construct a boxplot [3 marks]
- Construct a 95% confidence interval for the population mean [2 marks]
- How large a sample do we need to estimate the population portion with a margin of at most 0.4 units with 95% confidence? [2 marks]
- Test $H_0 : \mu = 6$ against $H_A : \mu \neq 6$ at 5% level of significance [2 marks]
- Test $H_0 : \mu = 6$ against $H_A : \mu > 6$ at 5% level of significance [1 marks]. Given $Z_{0.975} = 1.96$ and $Z_{0.95} = 1.645$

Que:6 Suppose a discrete random variable X is the outcome of a three-faced dice (sample space of X is $\Omega \equiv \{0, 1, 2\}$). The PDF of X is given by:

$$\Pr\{X = 0\} = \frac{1}{5}, \quad \Pr\{X = 1\} = \frac{3}{5}, \quad \Pr\{X = 2\} = \frac{1}{5}.$$

(a) If this dice is rolled $n = 90$ times, find $\mathbb{E}[M_n]$ and $\text{Var}[M_n]$, where

$$M_n = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

and X_k is the k^{th} trial of the random variable X .

(b) Evaluate the quantity

[2 marks]

$$\frac{1.103 - \mathbb{E}[M_n]}{\sqrt{\text{Var}[M_n]}}$$

for finding the probability that $\Pr\{M_n > 1.103\}$ using the Central Limit Theorem (CLT). Given $Z_{0.975} = 1.96$ and $Z_{0.95} = 1.645$. [3 marks]

Que:7 Consider a random process $X(t) = At + b$, where A is uniformly distributed random numbers in the interval $[0, 2]$, and b is constant. Now, (a) sketch the ensemble (a few waveforms) of the random process, (b) find expectation of random process $E[X(t)]$, and (c) check whether $X(t)$ is a wide-sense stationary (WSS) process or not. [5 marks]

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